

CoRoT-2b

R-0169

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_180725 · created 2026-07-16T17:30:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458324.8599 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.064
Transit depth (Rp/Rs)^2	4.98 +/- 2.86 [%]
Orbital Inclination (inc)	89.95 +/- 2.07
Airmass coefficient 1 (a1)	0.51 +/- 0.34
Airmass coefficient 2 (a2)	0.31 +/- 0.3
Scatter in the residuals of the lightcurve fit is	64.23 %
Best Comparison Star	None
Optimal Aperture	5.36
Optimal Annulus	6.09
Transit Duration (day)	0.0999 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.064 vs 0.1667 (deviation 0.44σ)
Duration fit vs published	0.0999 d vs 0.095 d (deviation 0.41σ)
Mid-time O–C	-6.5 min
Depth SNR	1.8
Residual scatter	64.23 %

Light curve

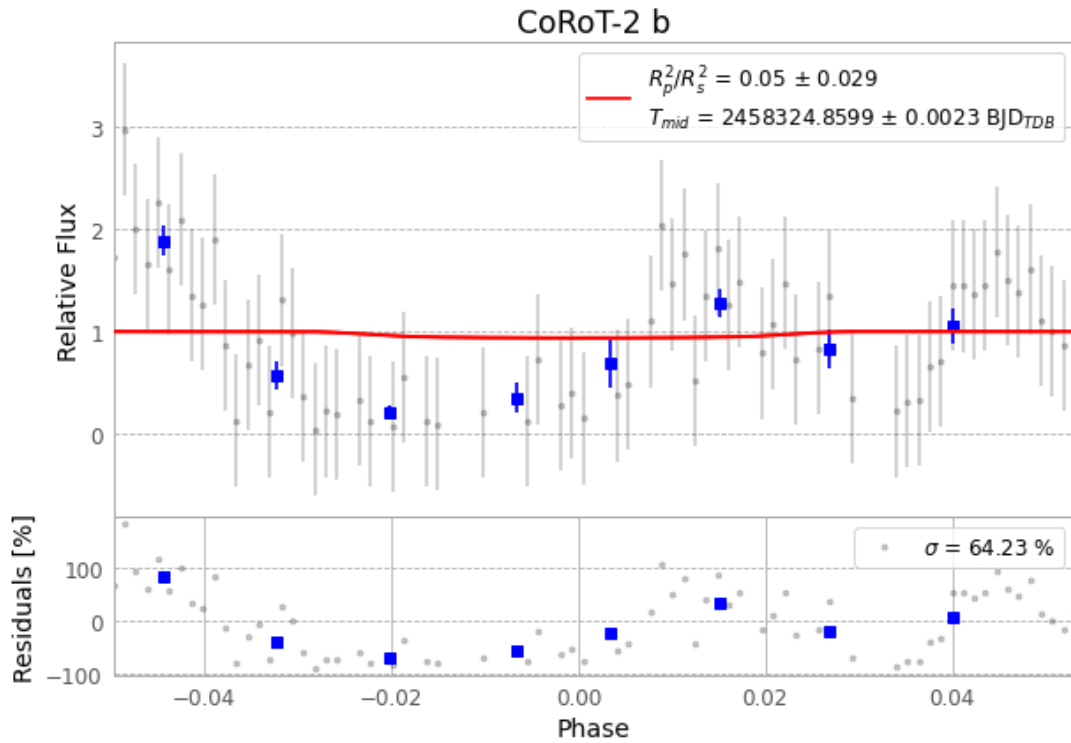


Figure 1 — FinalLightCurve_CoRoT-2 b_2018-07-24.png (SHA-256 fe9d4f53139d56c5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0b1747cb4901cbdf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62be8a4

Data provenance

87 FITS frames (57 MB), CoRoT-2180725062412.FITS → CoRoT-2180725104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180725.log (ee36afa6e1ce...), FinalLightCurve_CoRoT-2 b_2018-07-24.png (fe9d4f53139d...), FinalLightCurve_CoRoT-2 b_2018-07-24.csv (572448af273a...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:30Z.